Group Coursework

This is a group coursework (minimum 4 students, maximum 5 students in each group). The coursework has two components: i) an implementation component (70%), and ii) a related vulnerability analysis component (30%) and. **The coursework (both components) will be marked out of 100% which carries 100% of the overall grade.**

i) Implementation part - Company scenario

The objective of the implementation component is to give students an experience in dealing with a realistic network design problem with security features. This will give the opportunity to use relevant application software to implement a design scenario and address the relevant design issues in tackling a network related project in real life environment. The outline of the scenario is given below.

Network Scenario

You have been recently graduated and have been employed by the NSP Ltd company. The company has been commissioned to advise on the LAN requirements of a medium-sized company, MSS Ltd, that is composed of three departments: marketing, sales, and general support department. Marketing occupies the first floor, while sales and general support are on the second floor and third floor. Marketing has 20 workstations, sales has 25 workstations, and general support has 15 workstations. Some applications require that data be transferred between departments, but generally each department has its own applications. Everyone needs access to the Internet, the company's internal Internet (intranet), and e-mail.

In addition, each department has its own database of local information. The company wants to let all the employees from around the company access any of these databases. A backup database server is also needed.

Design a local area network solution for MSS Ltd to suit their present and future needs. Show the locations of all workstations, servers and interconnecting devices. Give justifications for your design, especially the type of local area network wiring, type of devices used, type of local area network topology and protocol. Include aspects of security issues that will be provided as a part of the proposed design. Show the design plan in details for the proposed network. Give as much details of your network design as it is possible and appropriate.

Use the following assumptions:

1. Appropriate software applications and large data sets reside on departmental servers.
2. To support e-mail and communication, a server is needed.
3. To support the company intranet, a server is needed.
4. To support database and backup database, servers are needed.
5. For backup and archive, a backup server is needed.
6. Additional redundancy plan should be considered in case of system failure.
7. Security issues –vulnerability analysis and maintenance issues need to be considered.
8. The company does not have an unlimited budget but is willing to invest in quality technology.

Requirements for the implementation part– a guideline

This assignment is a network design course work. There is no model answer for such design problem. The usual approach to such design problem is to assess the present scenario, and then to establish the requirements. Once the requirements have been established, a design plan is proposed and the design proposal is justified with appropriate explanation. The design can be implemented using some appropriate software, such as, **Packet Tracer or River Bed software,** and a future update proposition for the proposed design is suggested. **Finally, a guideline to carry out an appropriate vulnerability and security analysis of the proposed design using appropriate tools or application software need to be given.** Based on these criteria, the implementation report is expected to have the following sections as given below.

1) Introduction, aim, and overview of the present scenario. (5%+5%+5%=15%)
2) Analysis of the present scenario, and the requirements of the company. (15%)
3) Proposed LAN design - this section should include the detail building/room planning and the layout for the proposed LAN design. This can be expanded as outlined in (4). (15%)
4) Details of the proposed LAN design - the section can have the following sub-sections:
   a) Topology/Architecture (6%)
   b) Media/ Cabling (6%)
   c) Interconnectivity of networks between different sections and floors (6%)
   d) Outside connections (6%)
   e) Protocol/software/security or any other relevant issues, (4%) and
   f) Design implementation and any other additional issues. (12%)
5) Future proposition, and summary/conclusion. (7%+8%=15%)

- Total – 100%

  Detailed justifications and explanations are expected at every stage of the proposed implementation work.

Any general information for section (3), (4) and (5) will be considered as **'unnecessary'**. The **'general information'** can be categorised as any of the following items:

i) General textbook definitions of different types of networks.
ii) General textbook definitions of different types of media/cable.
iii) Any other general textbook definitions, such as, protocols, SNMP, etc.
iv) Any general type of diagram.
v) Definition of devices used and specially in the case of branded products.
vi) General information on Firewall, Virus software, and other access control rules.

All these types of general information can be included in the Appendix as 'Additional Information' if needed. Any information taken from a website without any modifications will also be considered as 'general information'. Details of any devices, if necessary, can be given in the Appendix. **References must be given if any information has been taken directly from any website/book/article.**

Marking scheme based on three major categories

a) Introduction/Assumption, overview, and analysis (IOA - 30%) - this includes part (1) and part (2) as outlined above.

b) Design, justification/explanation, and implementation (DJI - 55%) - this includes part (3) and part (4) as outlined above.

c) Overall presentation, future proposition, and conclusion (OPFC - 15%) - this includes part (5) as outlined above.

- Total = 100%

Percentage of this part of the report is 70%.

ii) Vulnerability analysis on Network Scenario

The main objective of the **vulnerability analysis** is to investigate security vulnerabilities that exist in a network environment. This will give you the scope to investigate in more depth security threats that are widely experienced in a network environment. The vulnerability analysis will lead the investigation to find out how a given network can be compromised and bypassed. The investigation will explore the security aspects of a given network through penetration test. It will investigate how to carry out a penetration test using various tools to find the vulnerabilities and assess the risk associated with potential security breaches.

Such type of vulnerability analysis can be carried out by using different types of tools, such as Kali Linux or Windows based tools. Kali Linux provides various tools for the following tasks: i) information gathering, ii) sniffing and spoofing, iii) vulnerability analysis, iv) web application analysis, v) password attacks, vi) exploitation tools, vii) wireless attacks, etc.

Your **vulnerability analysis** report should investigate in detail the following issues:

i) Security vulnerabilities in the given network setup, (20%).
ii) Kali Linux/Window tools that are suitable to carry out all the relevant vulnerability analysis as discussed above in such a given network, (40%).
iii) Defence mechanisms that can be taken as a result of the findings from the vulnerability analysis test, (30%), and
iv) Guideline for using suitable security and vulnerability analysis as a part of regular maintenance activities in such given network (10%).

Your investigation should give implementation examples for each relevant tool to show how the relevant tools can be used within a given scenario.

An activity list report needs to be attached at the end of the report in the Appendix which will outline the contribution of each member in the group.

- Total = 100%

Percentage of this part of report is 30%.

The submission deadline for this group cw report is **11 December 2026**. The final report should not exceed more than **20 PAGES.** Diagrams can be given in the Appendix if needed.

Late submissions will be decremented by 5% for every day the report is overdue to the maximum of 25% of the overall marks. Late submissions with valid extenuating circumstances must be notified to the students' office prior to the deadline for submission of coursework.